

HUMOS – World2Data: Navigation Ground Truth from Video

HUMOS is an AI-powered perception pipeline that converts raw video into structured, navigation-relevant ground truth for humanoid robots. It detects and tracks objects across frames, estimates object/navigation state (for example open/closed/ajar/blocked), detects hand-object interactions, and exports temporal annotations with evaluation metrics.

What HUMOS Does

- Object detection + persistent tracking IDs across time
- Segmentation for precise object boundaries (SAM3, FastSAM, RF-DETR Seg)
- Zero-shot state classification (SigLIP) for fast open/closed-style labels
- VLM reasoning (LFM2.5-VL backends) for richer semantic state/affordance output
- Temporal change detection and navigation timeline generation
- Ground truth JSON export and accuracy evaluation against manual labels
- Live webcam inference with auto-record + post-stop analysis
- Per-stage timing breakdown for performance profiling

Core UX

HUMOS provides two Streamlit workflows:

- 1) Upload workflow (``app.py``): run full multi-model analysis and inspect synchronized tabs
- 2) Live workflow (``pages/live.py``): real-time inference overlay, auto-recording, then full post-analysis on stop

Views include: Original, Combined, Tracking, Detections, Segmentation, Interactions, VLM Output, Temporal Changes, Ground Truth JSON, and Accuracy.

Technical Stack

Python, Streamlit, OpenCV, PyTorch, Ultralytics YOLO (v8/v11), RF-DETR, FastSAM, SAM3, MediaPipe, SigLIP, and LFM2.5-VL (Transformers/ONNX/MLX).

Quick Start

```
uv sync
streamlit run app.py
```

If ``uv`` is unavailable:

```
python -m venv .venv
source .venv/bin/activate
pip install -e .
streamlit run app.py
```

Impact

HUMOS bridges perception and robot actionability: it transforms raw video into machine-usable temporal knowledge for planning, imitation learning data generation, behavior debugging, and navigation/manipulation evaluation.